

Title: Intelligent Officer Workload Balancing using Bipartite Matching (Hungarian Algorithm)

1. Problem Statement

In traditional law enforcement, assigning Investigating Officers (IOs) to First Information Reports (FIRs) is a manual, biased, and inefficient process. This leads to severe officer burnout, skill mismatches (e.g., assigning financial fraud to a narcotics officer), and geographic inefficiencies. JusticeFlow solves this by treating case assignment as a mathematical Optimization Problem (Minimum-Cost Bipartite Matching).

2. The Cost Matrix (Feature Engineering)

Before applying the algorithm, the AI agent constructs an

$N \times M$

$N \times M$

Cost Matrix (

N

N

Cases,

M

M

Officers). The "Cost" of assigning an officer to a case is calculated using four heavily weighted penalties. Lower scores represent better matches.

- **Workload Penalty (Weight: 35%):** Uses an exponential penalty formula $(\text{active_cases} / \text{MAX_CASES})^2$. An officer with 9 cases receives an 81% penalty, protecting them from burnout.
- **Skill Mismatch (Weight: 30%):** Matches the Case Crime Type to the Officer's Qualifications. Exact match = 0.0 penalty. Partial match (General Investigation) = 0.5. No match = 1.0 penalty.
- **Rank Suitability (Weight: 20%):** Maps Case Severity to CrPC Rank Requirements (e.g., CRITICAL cases require DSP or above). Officers below the required rank receive a maximum 1.0 penalty.

- **Geographic Distance (Weight: 15%):** Calculates the Haversine great-circle distance between the Officer's station centroid and the crime scene's latitude/longitude, normalized over a 50km radius.

3. The Algorithm Selection

We utilize the **Hungarian Algorithm** (implemented via `scipy.optimize.linear_sum_assignment`).

- **Why?** Unlike greedy algorithms that make locally optimal choices, the Hungarian algorithm guarantees the global mathematical minimum cost assignment in
- $O(n^3)$
- time complexity. It elegantly handles rectangular matrices where the number of cases does not equal the number of officers, leaving the least optimal cases unassigned for the next cycle.

4. System Integration (OS & Database)

The AI agent operates as an isolated OS process spawned by the C++ Scheduler.

- **Database Isolation:** It reads unassigned cases from the `public` schema but only writes suggestions to the `analytics.Officer_Workload_Assignments` table, ensuring AI predictions never corrupt operational police data.
- **Inter-Process Communication (IPC):** Upon completion, the Python agent packs its execution metrics into raw C-bytes using `struct.pack` and writes them via non-blocking POSIX syscalls (`os.O_NONBLOCK`) to a Named Pipe (`/tmp/justiceflow_workload.fifo`) and POSIX

Shared Memory (`/justiceflow_shm`). This allows the C++
Dashboard to display real-time AI health without database querying.